

Mid Suggestion And Answers

CHAPTER 1

1. What is MIS?
2. Functions of MIS
3. Dimensions of Information System
4. Objectives of MIS, From the examples identify and explain the objectives

CHAPTER 2

1. Business Process, explain with an example involving a department
2. How Information Technology improves Business Processes
3. Systems for Different Management Groups, understand **TPS, MIS, DSS, ESS**, their differences and purposes
4. Enterprise Applications, understand their purpose and **ERP, SCM, CRM, KMS**
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CHAPTER 3

1. Organization, definition and common features
2. Porter's Five Competitive Forces
3. Porter's Generic Strategies

Chapter 1

Management Information Systems

A Management Information System (MIS) is a computer-based system that collects, processes, stores, and distributes information to support communication, automation, business processes, decision making, and coordination across an organization. It gives managers at every level, strategic, tactical, and operational, the right information, in the right form, at the right time, to run and control the business effectively.

- 1. Collection of Data:** MIS collects data from different business activities such as sales, employees, customers, and transactions.
- 2. Processing of Data:** The collected data is classified, arranged, calculated, and processed to make it meaningful.
- 3. Storage of Information:** MIS stores processed information so that it can be accessed and used when required.
- 4. Providing Useful Information:** MIS provides information in the form of reports, summaries, charts, and other useful outputs.
- 5. Supporting Management:** It provides managers with the right information at the right time to support decision making, coordination, and control.

Strategic Objectives

Information systems are essential because modern businesses depend on accurate and timely information to operate effectively. They help organizations achieve several important strategic objectives:

- 1. Operational excellence:** Improving efficiency and productivity to raise profitability.
- 2. New products, services, and business models:** Use technology to create new ways of doing business.
- 3. Customer and supplier intimacy:** Understand and serve customers and suppliers better.
- 4. Improved decision making:** Provide accurate and timely information instead of relying on guesswork.
- 5. Competitive advantage:** Help firms perform better than competitors.
- 6. Survival:** Help businesses meet technological and legal requirements necessary to continue operating.

Therefore, information systems are not simply computer systems. They are important business tools that help organizations operate, manage resources, make decisions, and achieve strategic objectives.

Functions of information system

An **Information System (IS)** is a set of interrelated components that **collect, process, store, and distribute information** to support communication, automation, business processes, decision making, and coordination across an organization. An IS combines technology with people and business processes.

An information system works through three basic functions, plus a fourth one that keeps it accurate:

1. **Input:** Collects raw data, such as customer orders, sales transactions, or employee attendance.
 2. **Processing:** Classifies, arranges, and calculates the data to transform it into useful information.
 3. **Output:** Produces useful results such as reports, summaries, charts, or documents.
 4. **Feedback:** Sends output back to the system to check and correct input when necessary.
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Dimension

Beyond the technical process, the information system has three major dimensions. Understanding these three dimensions provides **information systems literacy**, which blends behavioral and technical perspectives rather than focusing solely on computer hardware.

1. **Organization:** Includes people, organizational structure, business processes, politics, and culture. It determines how work and responsibilities are distributed, organized and to be performed.
2. **Management:** Includes decision making, problem solving, resource allocation, leadership, and strategy. Managers use information systems to address business challenges and solve them using tools.
3. **Technology:** Includes computer hardware, software, data management technology, and networking and telecommunications technology. It acts as the physical and digital backbone.

Thus, an effective information system requires the proper combination of **organization, management, and technology**.

Chapter 2

Business Process

A business process is a **structured, repeatable sequence** of coordinated tasks and activities designed to achieve a specific organizational goal or produce a consistent outcome.

- 1. Collection of Activities:** A business process consists of several activities performed to achieve a particular business result.
- 2. Coordination of Work:** Different employees or departments coordinate their activities to complete the process.
- 3. Flow of Information:** Information moves between employees, departments, customers, and business partners.
- 4. Cross-functional Nature:** Some business processes involve more than one department.

For example, fulfilling a customer order may require coordination among **sales, accounting, manufacturing, delivery partners, and the customer**. Information systems allow information to flow rapidly between these participants.

Business Process = How work gets done

How IT Improves Business Process

Information technology improves business processes by making work faster, easier, more efficient, and better coordinated. It can automate existing activities and also create new ways of doing business.

- 1. Automation:** IT can automate manual activities such as checking customer credit, generating invoices, and preparing shipping orders.
- 2. Faster Information Flow:** Information can be shared quickly between employees and departments.
- 3. Simultaneous Activities:** IT can allow activities that previously happened one after another to occur simultaneously.
- 4. Reduced Decision Delays:** Managers can receive information quickly and make decisions faster.
- 5. Creation of New Business Models:** Technology can create new activities and business models, such as online shopping, e-books, and music streaming.

Therefore, information systems improve business processes by making them **faster, more efficient, coordinated, and responsive**.

Information System = Helps information flow through that work

5. How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?

Systems for Different Groups

Organizations have different management levels, so they use different information systems.

1. Operational Management

Operational managers handle **day-to-day activities and routine decisions**. They mainly use **Transaction Processing Systems (TPS)**.

TPS records routine transactions such as sales, payroll, shipping, reservations, and employee records. Its information must be current, accurate, and easily available.

2. Middle Management

Middle managers are responsible for **monitoring, control, and short-to-medium-term decisions**. They use **Management Information Systems (MIS)**.

MIS summarizes data from TPS and produces routine reports that help managers monitor current performance and control the business.

3. Senior Management

Senior managers focus on **strategic decisions and long-term planning**. They use **Executive Support Systems (ESS)**.

ESS provides summarized internal and external information through dashboards and other easy-to-use interfaces. It helps senior managers make nonroutine decisions requiring judgment and insight.

Decision-Support Systems (DSS): a computer program that helps leaders solve complex, changing problems where the right answer is not known ahead of time.

Very important exam structure:

Operational → TPS → Routine transactions
Middle → MIS → Monitoring and control
Senior → ESS → Strategy and long-term decisions
DSS → Decision-Support Systems

Enterprise Application

Enterprise applications connect different functional areas and management levels. They improve organizational performance by integrating business processes and allowing information to flow across departments.

The four major enterprise applications are:

1. **ERP:** Integrates major internal business functions such as manufacturing, finance, sales, marketing, and human resources into one system.
2. **SCM:** SCM manages relationships with suppliers, distributors, purchasing firms, and logistics companies.
3. **CRM:** CRM manages customer relationships by coordinating sales, marketing, and customer service activities.
4. **KMS:** KMS helps organizations collect, store, share, and apply knowledge and expertise.

In conclusion, enterprise applications integrate different business activities and help organizations improve efficiency, coordination, customer service, and decision making.

Very important exam structure:

Enterprise Resource planning	→	→	ERP	→	Internal functions
Supply Chain Management	→	→	SCM	→	Suppliers
Customer Relationship Management	→		CRM	→	Customers
Knowledge Management Systems	→		KMS	→	Knowledge

6. What is the role of the information systems function in a business?

The **information systems function** is responsible for managing and maintaining the technology and information systems that support business operations.

In most organizations, the **Information Systems Department** is responsible for IT services, including:

- Hardware
- Software
- Data storage
- Networks
- IT infrastructure

Important IS specialists include:

1. **Programmers:** Write software instructions (codes) for computers.
2. **Systems Analysts:** Connect the IS department with the rest of the organization and translate business requirements into information systems.
3. **Information Systems Managers:** Manage programmers, analysts, projects, facilities, telecommunications, databases, and computer operations.
4. **Chief Information Officer (CIO):** Oversees the organization's use of IT and aligns technology with business strategy.
5. **Other senior roles** may include **CSO/CISO, CPO, CKO, and CDO**, responsible respectively for security, privacy, knowledge management, and data governance.

The IS function has also evolved from mainly performing technical programming tasks to acting as a **strategic business partner and change agent**. IS professionals can propose new strategies, information-based products, services, and business improvements.

Finally, **IT governance** establishes the strategy, policies, decision rights, and accountability for using IT so that technology supports organizational objectives.

Remember:

IS Function = Manage Technology + Support Business + Align IT with Strategy

1. Chief Information Officer = CIO
2. Chief Security Officer = CSO
3. Chief Information Security Officer = CISO
4. Chief Privacy Officer = CPO
5. Chief Knowledge Officer = CKO
6. Chief Data Officer = CDO

Chapter 3

10. Define Organization and Explain the Common Features of Organizations.

Introduction:

An organization is a stable, formal social structure that takes resources from the environment and processes them to produce outputs. Technical and behavioral perspectives are two ways to define an organization.

Common features:

1. **Routines and Business Processes:** Routines are established rules and practices employees follow. A collection of routines forms business processes.
2. **Organizational Politics:** Different people have different interests and perspectives, which can create competition and conflict over resources and decisions.
3. **Organizational Culture:** Culture consists of shared beliefs and assumptions about how an organization should work. It can encourage common practices but may also resist technological change.
4. **Organizational Environment:** Organizations depend on their environment for resources, customers, employees, and support. They also influence their environment.
5. **Organizational Structure:** Structure defines how responsibilities and authority are arranged. Different structures influence the information systems used by organizations.

Closing:

Therefore, managers must understand organizational features because they directly affect how information systems are designed, implemented, accepted, and used.

11. Explain Porter's Five Competitive Forces.

Introduction:

Michael Porter's Five Competitive Forces model explains the competitive environment of a business. It helps organizations understand the pressures that affect their competitive position and develop suitable strategies. MIS Chapter 3

Main Points:

1. Traditional Competitors:

Existing competitors try to attract customers through better products, pricing, branding, and services.

2. New Market Entrants:

New companies may enter an industry when barriers to entry are low. High capital or knowledge requirements can make entry difficult.

3. Substitute Products and Services:

Customers may switch to alternative products or services when they provide better value or lower prices.

4. Customers' Bargaining Power:

Customers have greater power when they can easily switch to competitors or compare prices.

5. Suppliers' Bargaining Power:

Suppliers have greater power when businesses have limited alternative suppliers.

Closing:

In conclusion, businesses need to understand these five forces to identify competitive threats and develop strategies for achieving competitive advantage.

12. Explain Porter's Four Generic Strategies.

Introduction:

Organizations can use different strategies to respond to competitive forces. According to the chapter, information systems can directly support four generic strategies for achieving competitive advantage. MIS Chapter 3

Main Points:

1. **Low-Cost Leadership:**

A company uses information systems to reduce operating costs and offer lower prices while maintaining quality and service.

Example: Walmart's continuous inventory replenishment system.

2. **Product Differentiation:**

A company uses information systems to create new products, services, or convenience that competitors find difficult to copy.

Example: Apple's iPod/iTunes and Nike's custom NIKEiD.

3. **Focus on Market Niche:**

A company uses information systems to analyze customer data and serve a specific market segment better than competitors.

Example: Hilton Hotels' OnQ guest-preference system.

4. **Customer and Supplier Intimacy:**

A company uses information systems to build strong relationships with customers and suppliers, increasing loyalty and making it harder to switch to competitors.

Example: Amazon's purchase recommendations.

Closing:

Thus, these four strategies help organizations use information systems to respond to competitive pressures and create competitive advantage.